Development of a Multi-Channel Field-Based Docking Framework for Efficient Virtual Screening

Scoring molecular binding as a field-overlap landscape, not a single number

Keywords: Virtual Screening · Protein-Ligand Docking · Fast Fourier transformation · ligand Torsion

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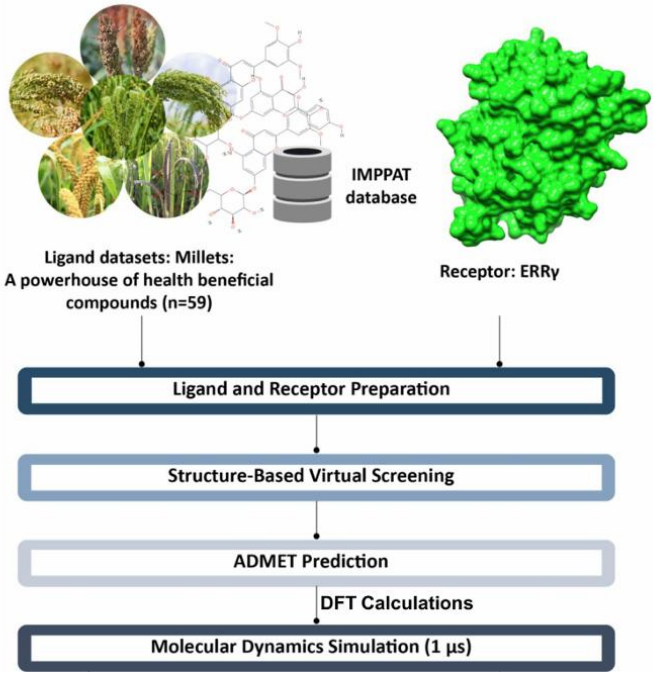
Functional Genomics & Bioinformatics Lab

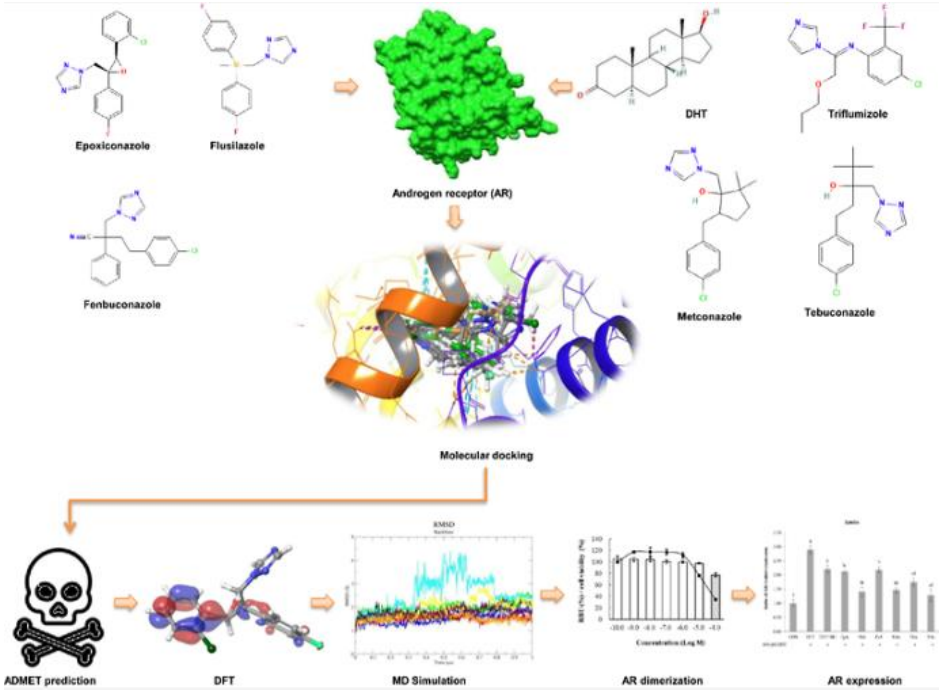
Index

01	Introduction	3-5
02	Objective	6
03	Materials and Methods	7-9
04	Results	10-13
05	Discussion	14
06	Further study	15

Introduction

Application of Molecular Docking

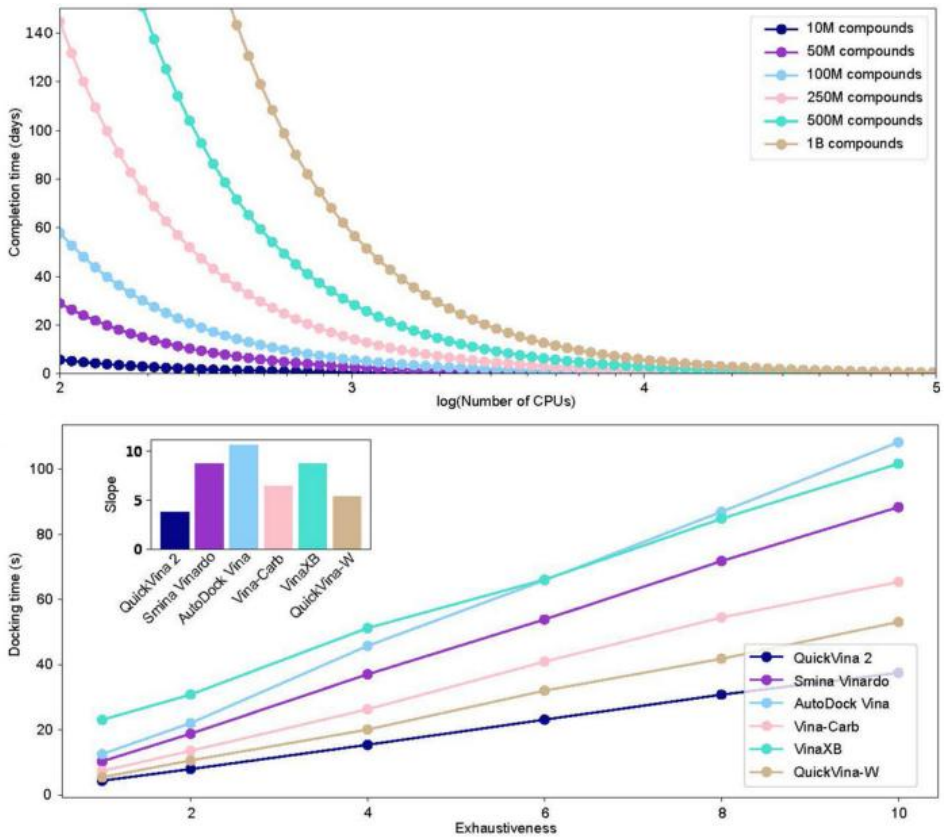




- Molecular docking supports drug discovery and molecular interaction studies
- It predicts binding poses and scores protein–ligand interactions
- Its computational cost limits large-scale virtual screening

Pathak, Rajesh Kumar, and Jun-Mo Kim. "A computational assay for identifying millet-derived compounds that antagonize the interaction between bisphenols and estrogen-related receptor gamma." Frontiers in pharmacology 15 (2024): 1435254.
Pathak, Rajesh Kumar, et al. "Exploring the Androgen Receptor Binding Affinity of Azole Derivatives through Multiscale Computational and Experimental Approaches." Environmental science & technology 59.45 (2025): 24323.

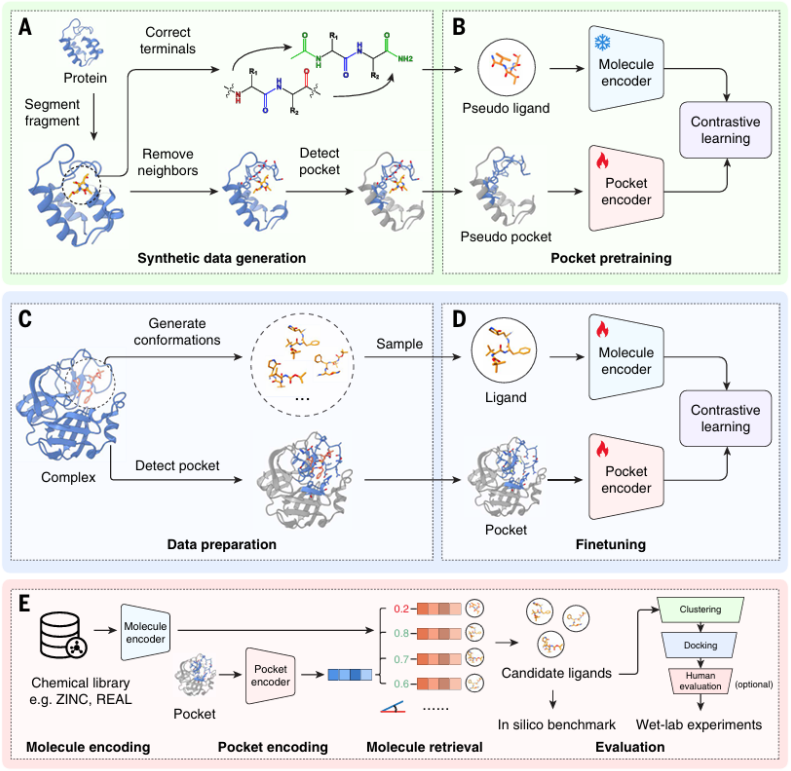
Computational Challenges in Large-Scale Docking

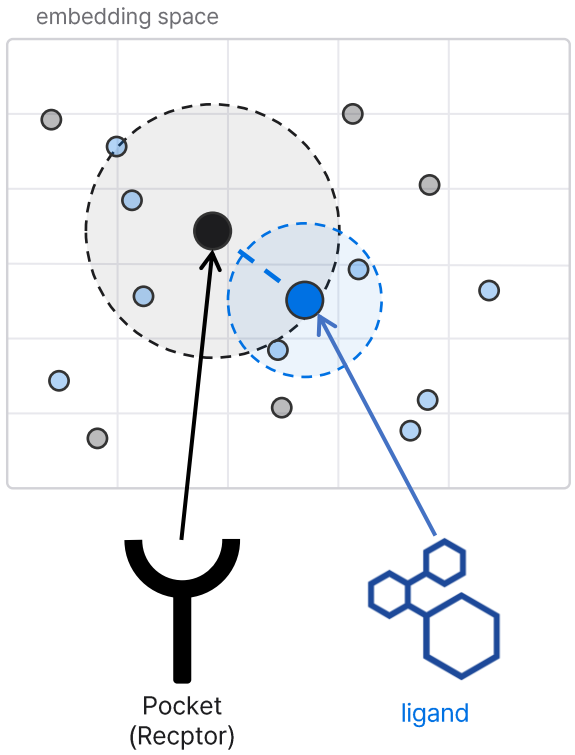


- Molecular docking remains computationally intensive
- Extending screening across protein targets increases computational requirements

Gorgulla, Christoph, et al. "An open-source drug discovery platform enables ultra-large virtual screens." Nature 580.7805 (2020): 663-668.
Jia, Yinjun, et al. "Deep contrastive learning enables genome-wide virtual screening." Science 391.6781 (2026): eads9530.

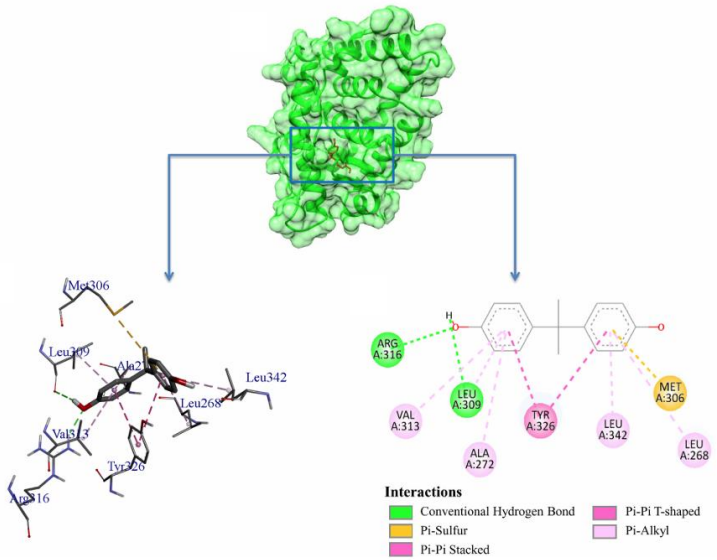
Contrastive Learning for Effective Virtual Screening

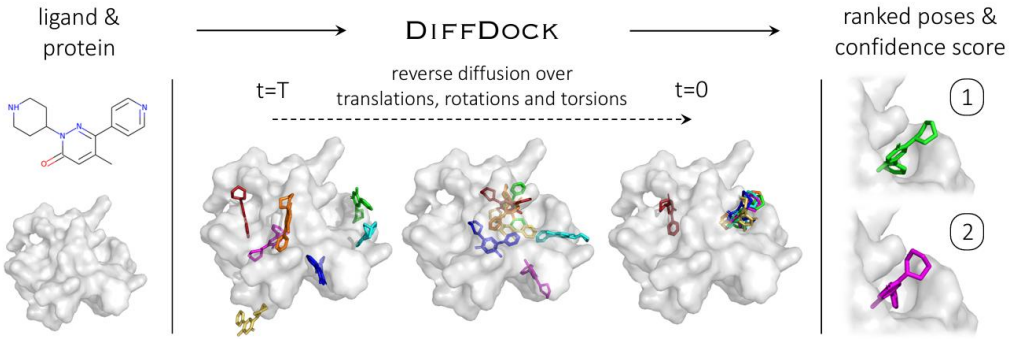




- Protein pockets and molecules are aligned in a shared embedding space.
- Embedding similarity enables rapid prioritization of candidate compounds.
- Precomputed embeddings support large-scale screening across protein targets.

Pose-Guided Learning for Candidate Retrieval

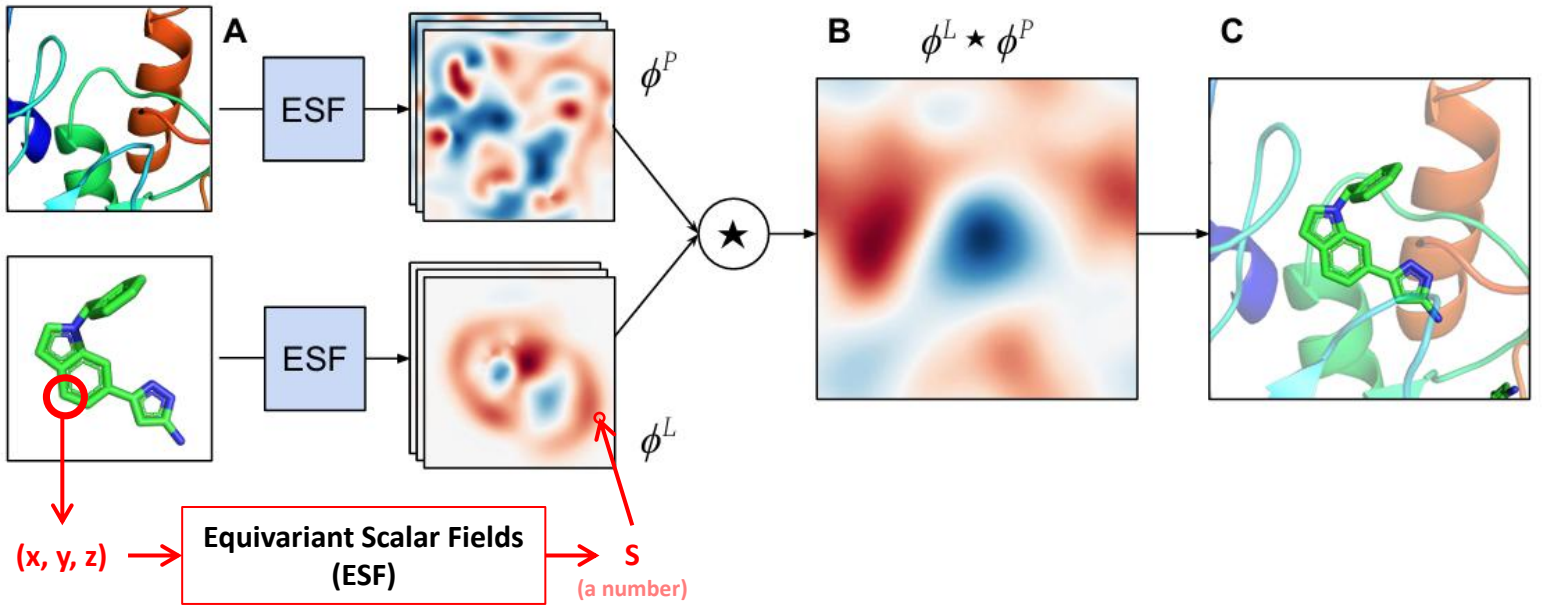




- Binding poses reveal structural interactions absent from retrieval scores
- **DiffDock** predicts poses through translational, rotational, and torsional updates
- Repeated evaluations introduce computational overhead at library scale.

Pathak, Rajesh Kumar, and Jun-Mo Kim. "A computational assay for identifying millet-derived compounds that antagonize the interaction between bisphenols and estrogen-related receptor gamma." Frontiers in pharmacology 15 (2024): 1435254.
Corso, Gabriele, et al. "Diffdock: Diffusion steps, twists, and turns for molecular docking." arXiv preprint arXiv:2210.01776 (2022).

FFT-Based Docking with Equivariant Scalar Fields

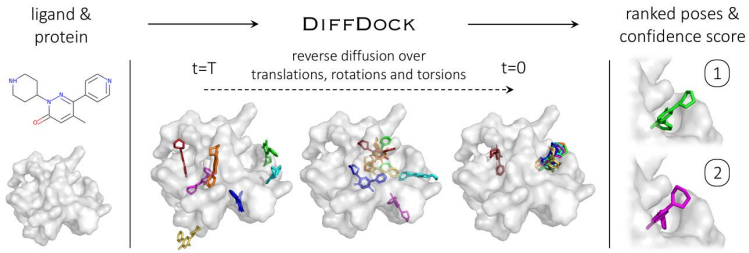


- FFT-based search efficiently optimizes ligand translation and rotation
- Internal conformations remain fixed within each rigid-body search
- Integrating torsional flexibility is a key step toward efficient flexible docking

Jing, Bowen, Tommi Jaakkola, and Bonnie Berger. "Equivariant scalar fields for molecular docking with fast fourier transforms." International Conference on Learning Representations. Vol. 2024. 2024.

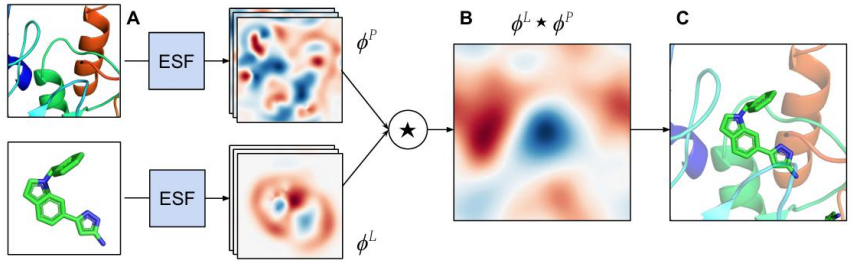
Integrating Search Efficiency and Ligand Flexibility

Pose Generation with Torsional Flexibility



- High computational cost
- Explicit torsional flexibility

FFT-Based Rigid-Body Search



- Fast pose search
- Fixed internal conformation

- There is a need for combining FFT-based search efficiency with ligand torsional flexibility

Jing, Bowen, Tommi Jaakkola, and Bonnie Berger. "Equivariant scalar fields for molecular docking with fast fourier transforms." International Conference on Learning Representations. Vol. 2024. 2024.

Research Question and Objective

Objective

To develop and evaluate an FFT-based docking framework incorporating ligand torsional flexibility

Question

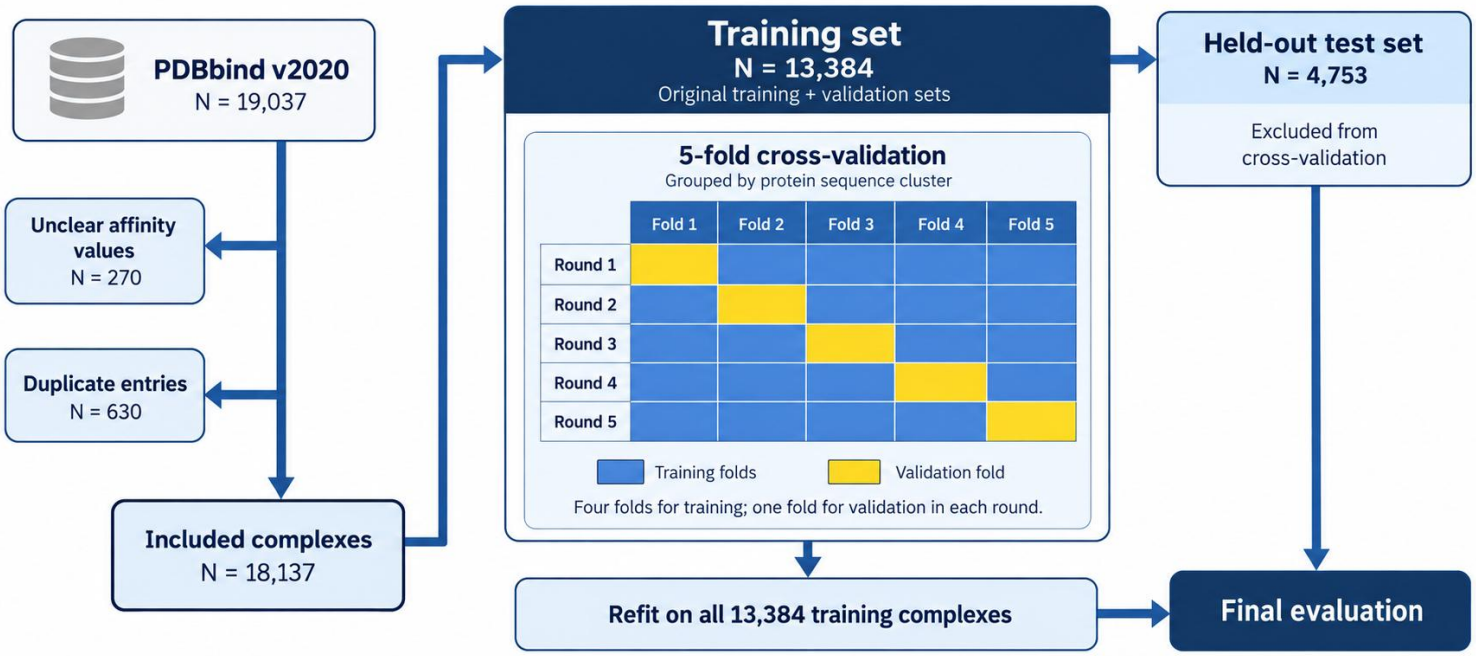
Can FFT-based docking account for ligand flexibility while maintaining computational efficiency?

Hypothesis

Incorporating torsional flexibility into FFT-based search can improve pose prediction and virtual screening accuracy while retaining computational efficiency

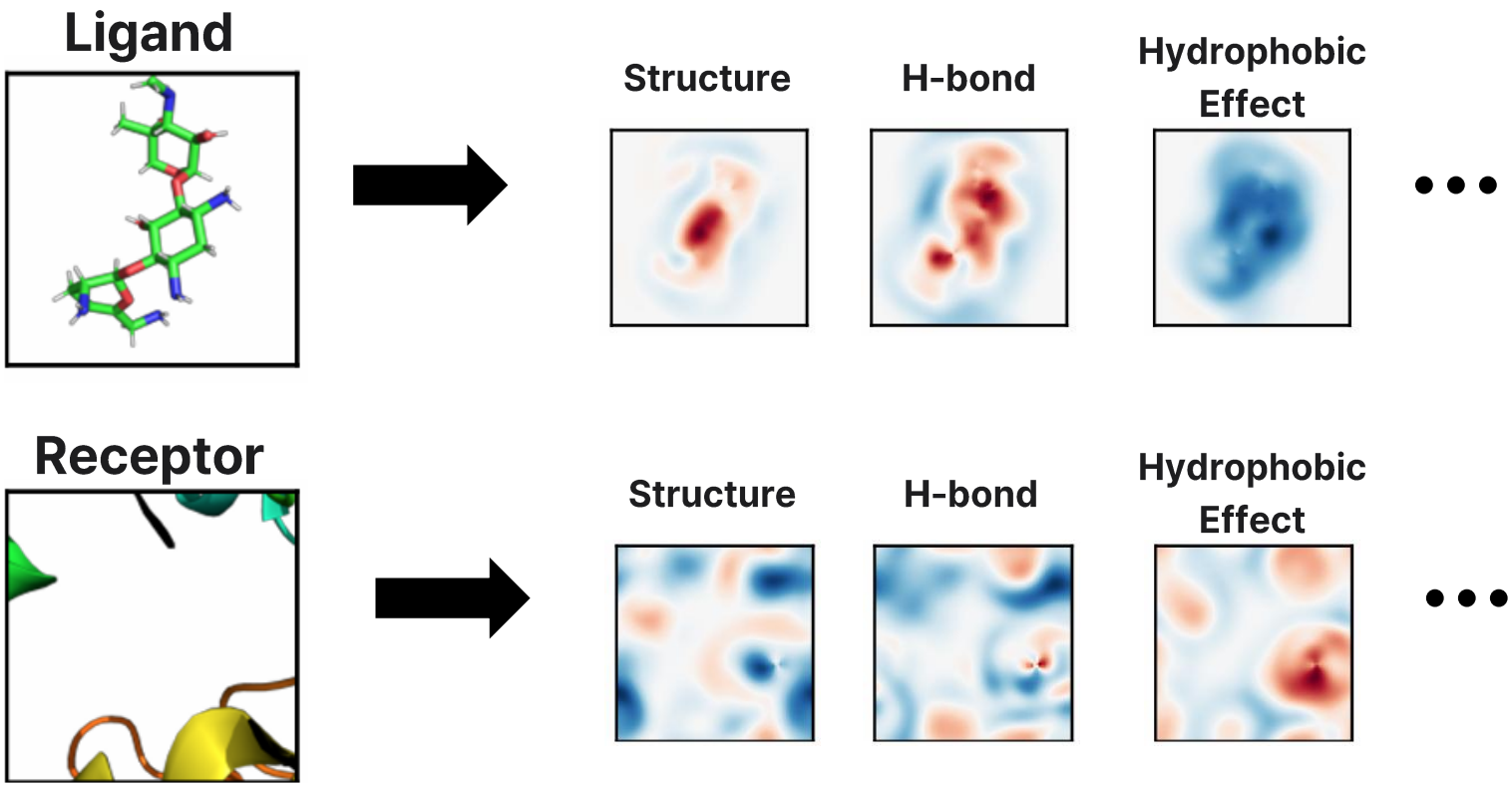
Materials & Methods

Data Preparation and Model Training/Test



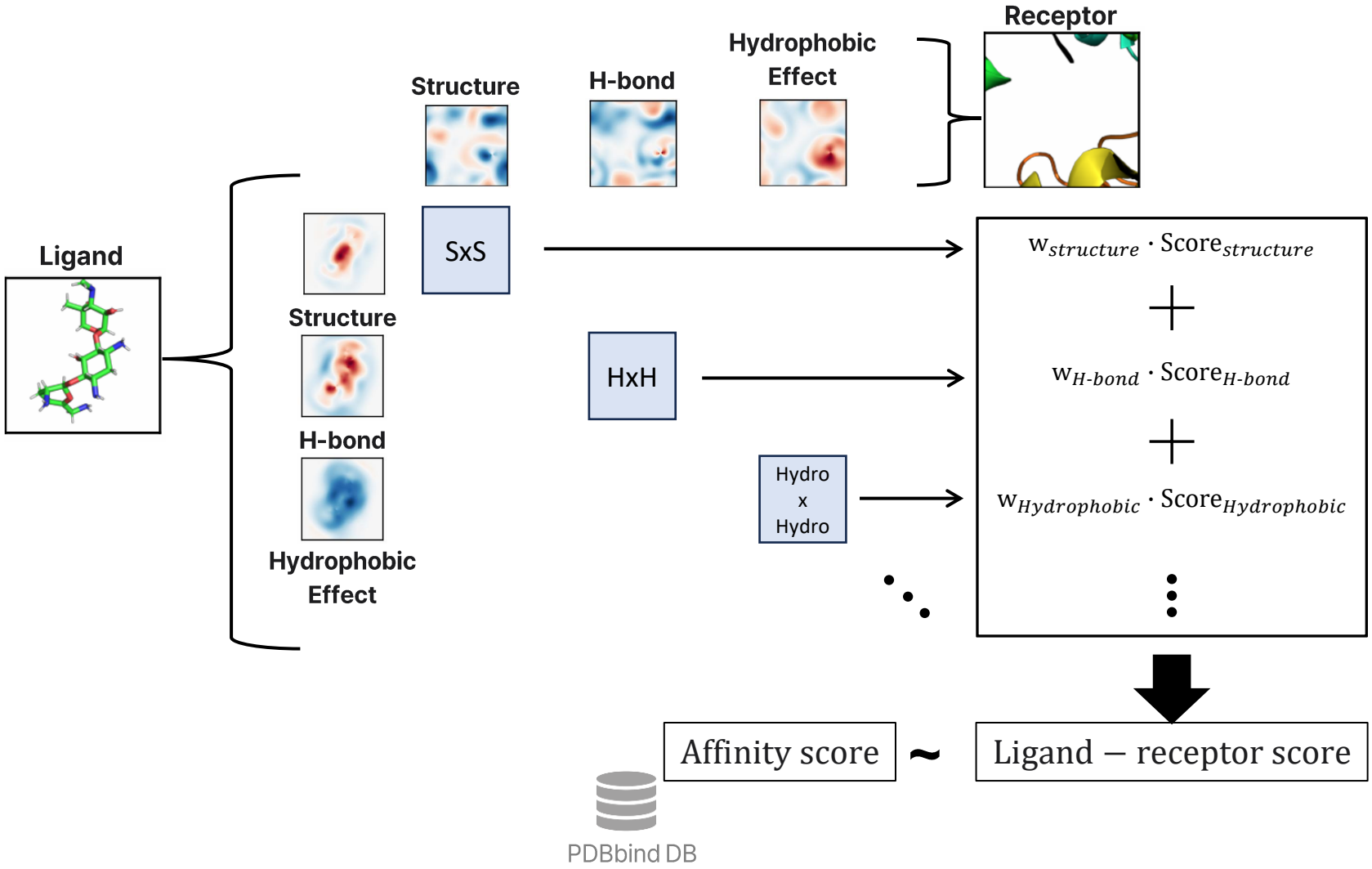
- Models were trained using five-fold cross-validation and evaluated on a held-out test set

Molecular Field Representation

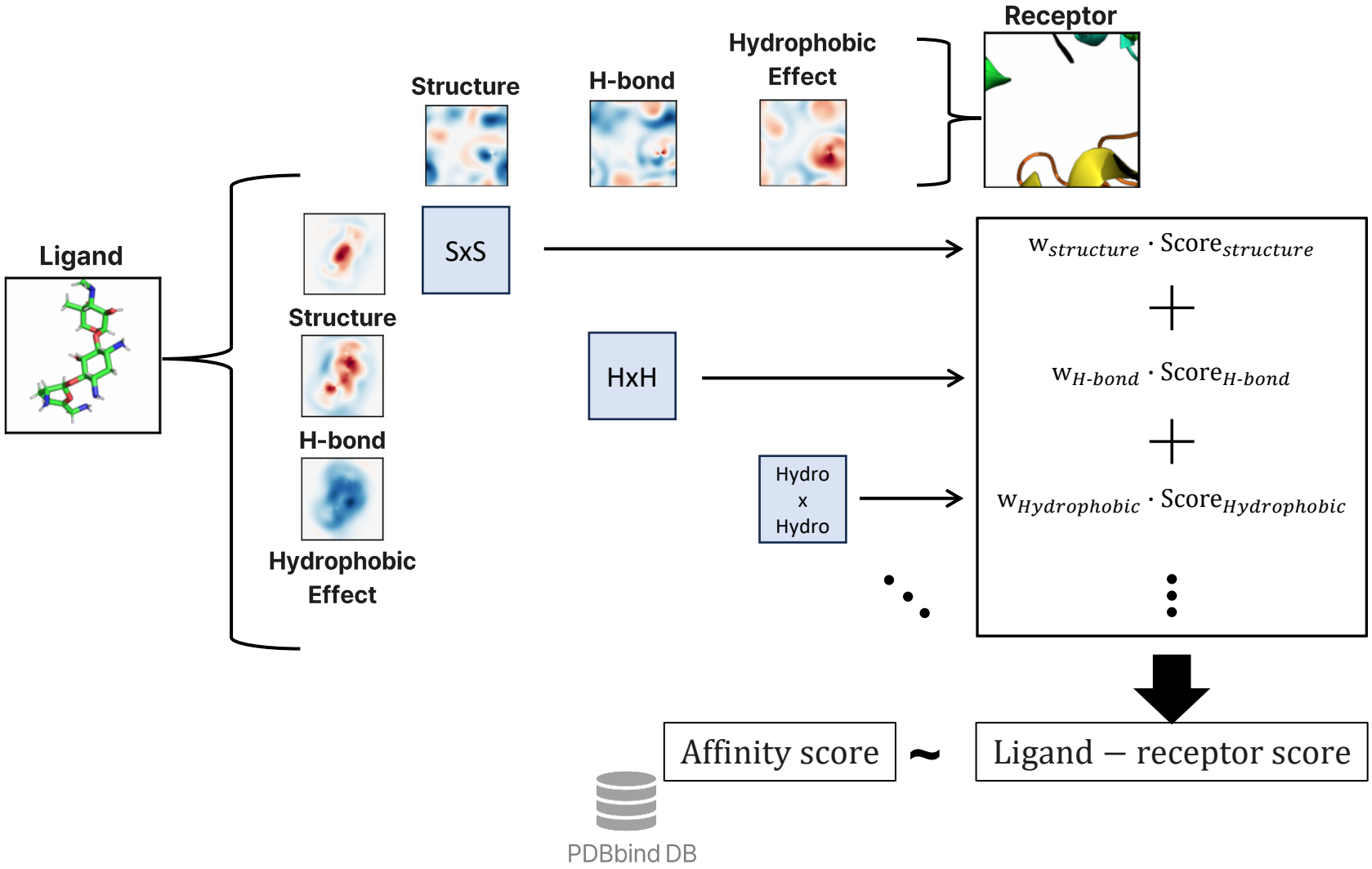


- Vina score $= \sum_{Steric,\, hydrophobic,\, hydrogen\text{-}bond} w \cdot \text{Score}$
- Ligand-receptor score $= \sum_{Vina\ terms,electrostatics,desolvation} w \cdot \text{Score}$

Learning Interaction Weights



Learning Interaction Weights



Data and evaluation

Rationale: a split that cannot leak, and two separate yardsticks

FIGURE

figures/07_data_cv.png

Sequence-clustered complexes assigned whole to folds, five-fold cross-validation, and a frozen held-out test set kept aside

RATIONALE

- PDBbind: 18,137 complexes kept from 19,037, clean '=' affinity entries only [1]
- Cluster-aware five-fold CV grouped by sequence, plus a frozen held-out test
- Affinity by Pearson r, Spearman ρ, RMSE; pose judged separately by native-pose recognition

Cluster splitting prevents memorising a protein family — the numbers that follow are meant to be honest, not flattering.

[1] PDBbind — Wang, Fang, Lu & Wang, J. Med. Chem. 2004, with subsequent database releases; complex counts above are from our own processing of the set.

RESULTS

Adding physics lifts the scorer

Preliminary, on the held-out PDBbind test set

FIGURE

figures/08_result_channel_climb.png

Test correlation rising as channels are added, from plain grid overlaps to the full channel set

step	channels	Ridge r	GBR r
grid overlaps	baseline	0.38	0.46
Vina-style pair terms	+contact / H-bond / entropy	0.43	0.49
+ electrostatics + desolvation	best so far	0.46	0.52

Held-out PDBbind test set; r is the correlation between predicted and experimental affinity at the crystal pose.

KEY POINTS

- Built up from shape overlaps to a full, Glide-like channel set
- Electrostatics and desolvation gave the single biggest lift
- Honest: still below a useful scorer (around r 0.7-0.8) — the point is the climb

Measured in this work on the held-out PDBbind test set; no comparison figures from other docking programs are quoted.

RESULTS

H-bond and electrostatics pay off in the pose

The same channels, weighted differently — and judged by the right yardstick

FIGURE

figures/09_result_pose_recognition.png

Native-pose recognition rate for three weightings of the same channels, shown as score-versus-RMSD funnels or as a bar comparison

weighting of the same channels	top pose < 2 Å
affinity-fit weights	2.4%
contact only	53%
physical pose weights	99.6%

n = 250 complexes. Decoys are rigid perturbations of the native pose with the native always present — a local funnel test, not blind docking.

KEY POINTS

- These channels barely move affinity r — affinity is the wrong yardstick for them
- Physically weighted, the same channels recognise the native pose almost always
- Lesson: ranking affinity and recognising a pose are two different jobs

Measured in this work (n = 250). The test is a local score-versus-RMSD funnel, stated here explicitly rather than reported as blind docking performance.

RESULTS

It binds like this; we predict that

Crystal pose against our top-ranked predicted pose, on held-out complexes

FIGURE

figures/10_result_pose_example.png

Two held-out complexes: the crystallographic pose and the top-scored predicted pose overlaid in the pocket, with RMSD labelled

KEY POINTS

- Crystal pose and our top-ranked predicted pose, overlaid in the same pocket
- Held-out complexes; the pose is picked by score alone, native never given
- The two nearly coincide: RMSD 0.87 Å and 0.77 Å

This is what the pose weights buy, shown on a complex the scorer never saw during training.

Measured in this work on held-out complexes; poses selected by score, with the native pose withheld from the selection.

RESULTS

The scoring funnel

Score against RMSD for three weightings of the same channels

FIGURE

figures/11_result_funnel.png

Score-versus-RMSD scatter for affinity-fit, contact-only and physical pose weights — a funnel narrowing towards low RMSD for the physical weighting

KEY POINTS

- A good scorer gives its highest scores to the lowest-RMSD poses
- Contact alone already funnels toward the native pose; physical weights sharpen it
- Affinity-fit weights do not funnel at all — the table's message, drawn

Optional slide: if time is short, the recognition table and the overlay already carry this point.

Measured in this work; same local-funnel setting as the pose-recognition test.

DISCUSSION

Limitations

What this work does not yet show

STATED PLAINLY

Scorer still weak
Affinity correlation is around r 0.5 at the crystal pose — below a useful scorer, and not yet at the level of classical docking.

A local funnel, not blind docking
Decoys are rigid perturbations with the native pose always present. True blind, global docking is untested.

Crude inputs
Gasteiger charges on PDB structures and name-based atom typing. Better force-field charges did not improve affinity r — an honest negative result.

Cognate redocking on PDBbind only
No external affinity test, and no enrichment (LIT-PCBA) or pose (PoseBusters) benchmark has been run yet.

None of these are hidden in the appendix: they are what the next phase of the project has to fix.

Further study

From a local funnel to blind docking, benchmarks, and an application

NEXT STEPS

Pose-appropriate weights
Learn field weights for the pose task rather than reusing affinity-fit weights, which the results show are the wrong ones for geometry.

Move to blind FFT docking
Leave the local funnel: search the full landscape over orientation and translation without the native pose present.

Run the held-out benchmarks
PoseBusters for pose quality (RMSD < 2 Å) and LIT-PCBA for enrichment (EF, BEDROC), neither of which has been run yet.

Apply to an animal-biotech target
Screen natural products (COCONUT) against TBK1 — aging and fat accumulation — recovering known binders such as betaine as a positive control. [1]

The order matters: fix the pose weights, then go blind, then benchmark, then apply.

Benchmarks: PoseBusters (Buttenschoen et al., Chem. Sci. 2024); LIT-PCBA (Tran-Nguyen et al., J. Chem. Inf. Model. 2020); COCONUT (Sorokina et al., J. Cheminform. 2021). [1] Betaine–TBK1 — reference to be confirmed by the author.

Docking is central — but slow

It answers whether and how a ligand binds, which is why it is worth fixing

CONCEPT FIGURE

docking = pose + affinity, but slow

A ligand docked in a pocket with its pose and affinity read out, next to a clock or a queue of compounds marking the per-compound cost

KEY POINTS

- Docking predicts whether and how a ligand binds — a core tool for finding bioactives [1]
- Crucially it returns the binding pose — the 3D conformer, not just a score
- But classical docking is computationally slow, and hard to run at library scale

Docking gives exactly what we want — the pose — at a cost that does not scale.

[1] Role of molecular docking in bioactive-compound discovery — reference to be confirmed by the author.

To go faster: DrugCLIP

Contrastive embeddings make screening fast

CONCEPT FIGURE

pocket and ligand as points, dashed line = similarity

Pocket vector and ligand vector in an embedding space, joined by a dashed line labelled 'similarity = one number'

KEY POINTS

- To screen faster, DrugCLIP embeds pocket and ligand as vectors; binding is approximated by similarity [1]
- Extremely fast over very large libraries — this is its genuine strength
- But it answers only 'binds or not': a single similarity score

The speed is real, and so is the price paid for it — which the next slide names.

[1] DrugCLIP, arXiv:2310.06367.

What is lost: the conformer

A score without a pose cannot be interpreted structurally

CONCEPT FIGURE

a score with no pose — pocket with a question-mark ligand

A pocket showing a high score but the ligand drawn as a question mark, beside the same pocket with an explicit docked conformer

KEY POINTS

- DrugCLIP gives no conformer — you cannot see how the ligand sits in the pocket
- It matters because classical docking does return that pose — what structural follow-up needs
- So speed came at the cost of the pose

> The question is not only which compounds bind, but in which conformer they bind.

BindCLIP: shows how it binds, but still no pose

A step toward interpretability that stops short of the conformer

CONCEPT FIGURE

BindCLIP — interaction hints, but no 3D pose

Pocket with highlighted interacting regions or an attention-style map, and the 3D conformer still shown as absent

KEY POINTS

- BindCLIP was proposed to indicate how a ligand binds [1]
- But it still does not return the actual binding pose
- The gap remains: fast methods cannot give the bound 3D conformer

Pointing at where the interaction happens is not the same as producing the conformer.

[1] BindCLIP — reference to be confirmed by the author.

A different representation: scalar fields, not points

Another line of work keeps the geometry a point discards

CONCEPT FIGURE

point → scalar field (or slides/assets/field_channels.png)

A molecule collapsing to one point on the left; the same molecule rendered as a smooth scalar field on a grid on the right

KEY POINTS

- Another line represents a molecule as a scalar field instead of a point [1]
- A field carries the geometry that a point throws away
- Binding is then scored by field overlap, with SE(3) / FFT correlation

A field moves with the molecule, so geometry survives the scoring step.

[1] Equivariant scalar fields (ESF), arXiv:2312.04323.

Its limitation: rigid only, no internal torsion

Rigid-body transforms cannot reach the flexible conformer

CONCEPT FIGURE

rigid body vs rotatable bonds (or slides/assets/fragment_tree.png)

The same ligand moved as a rigid block, beside the ligand with its rotatable bonds marked and bending into a different conformer

KEY POINTS

- That field method uses rigid-body transforms only
- So it cannot see internal torsion — the flexible rotatable bonds
- Real ligands flex, and rigid scoring misses the true bound conformer

This is the specific gap the project addresses: flexibility, inside the field framework.

Limitation of the rigid-body field formulation discussed above (ESF, arXiv:2312.04323).

Our move: redefine binding as a curve

A score landscape over pose, with torsion among its axes

CONCEPT FIGURE

point → curve (or slides/assets/point_vs_curve.png)

Left: a single score as a point. Right: a score curve over pose whose axes include internal torsion, with the peak marked as pose + score

KEY POINTS

- We redefine binding as a score curve over pose — a field-overlap landscape, not a single value
- The curve's axes include internal torsion, so we can see which flexible conformer binds
- Its peak gives the pose, and the FFT draws the whole curve cheaply

One object answers both questions: how strongly it binds, and in which conformer.

OBJECTIVE

Objective and hypothesis

Predict the pose — torsion included — and test whether that improves accuracy

CONCEPT FIGURE

torsion-aware pose prediction (or slides/assets/point_vs_curve.png)

The curve over pose with a torsion axis called out, leading to a predicted bound conformer in the pocket

GOAL · HYPOTHESIS · APPROACH

- Goal: predict not just whether a ligand binds, but in which pose — explicitly accounting for internal torsion
- Hypothesis: accounting for the docking pose (pose + torsion), not just an affinity score, improves accuracy
- Approach: field-overlap curve + FFT + torsion, validated on held-out data

The hypothesis is about the pose — so the results have to be judged on the pose, not only on affinity.

A molecule as a multi-channel field

Rationale: channels that mirror the physics of binding

CONCEPT FIGURE

multi-channel field (or slides/assets/field_channels.png)

Atoms rendered onto a grid as separate channels — contact/shape, clash, hydrophobic, H-bond, electrostatics, desolvation — combined by learned weights

RATIONALE

- Receptor and ligand rendered as fields on a grid, one channel per interaction type
- Channels mirror binding physics: contact and shape, clash, hydrophobic, H-bond, electrostatics, desolvation
- Score = weighted sum of channel overlaps; only the weights are learned, on experimental affinity

Keep the physics fixed and learn only the weights, so a good score still means a sensible pose.

Scoring the pose landscape by FFT and torsion

Rationale: why the whole curve, and why torsion belongs in it

CONCEPT FIGURE

FFT landscape / torsion tree (or slides/assets/corr_landscape.png)

A correlation landscape with its peak marked as the docked pose, beside a ligand cut into rigid fragments on a kinematic tree

RATIONALE

- One FFT gives the overlap at every translation for a fixed orientation; the max over orientation and translation is pose + score
- Internal flexibility: cut only bonds that move atoms, then score fragments on a kinematic tree
- FFT makes drawing the whole curve affordable; torsion is the part prior field methods missed

Translations come free per orientation — the budget goes to rotations and torsion, where the geometry is.

Data and evaluation

Rationale: a split that cannot leak, and two separate yardsticks

CONCEPT FIGURE

cluster-aware CV + held-out (or slides/assets/cv_split.png)

Sequence-clustered complexes assigned whole to folds, five-fold cross-validation, and a frozen held-out test set kept aside

RATIONALE

- PDBbind: 18,137 complexes kept from 19,037 — clean '=' affinity, leak-proof split, about 900 dropped [1]
- Cluster-aware five-fold CV grouped by sequence, plus a frozen held-out test — no protein leakage
- Affinity by Pearson r, Spearman ρ, RMSE; the pose is measured separately

Cluster splitting prevents memorising a protein family — the numbers that follow are meant to be honest.

[1] PDBbind — Wang, Fang, Lu & Wang, J. Med. Chem. 2004, with subsequent releases; the counts above are from our own processing of the set.

RESULTS

Adding physics lifts the scorer

Preliminary, on the held-out PDBbind test set

RESULT FIGURE

figures/channel_climb.png

Drop in as-is: test correlation rising as channels are added, from plain grid overlaps to the full channel set

step	channels	Ridge r	GBR r
grid overlaps	baseline	0.38	0.46
Vina-style pair terms	+contact / H-bond / entropy	0.43	0.49
+ electrostatics + desolvation	best so far	0.46	0.52

Held-out PDBbind test set; r is the correlation between predicted and experimental affinity at the crystal pose.

KEY POINTS

- Built up from shape overlaps to a full, Glide-like channel set
- Electrostatics and desolvation gave the single biggest lift
- Honest: still below a useful scorer (around r 0.7-0.8) — the point is the climb

Measured in this work on the held-out PDBbind test set; no figures from other docking programs are quoted.

RESULTS

The pose is where it pays off

Checking the hypothesis on the yardstick it was stated for

RESULT FIGURE

figures/pose_recognition.png

Drop in as-is: native-pose recognition for three weightings of the same channels

weighting of the same channels	top pose < 2 Å
affinity-fit weights	2.4%
contact only	53%
physical pose weights	99.6%

n = 250 complexes. Decoys are rigid perturbations of the native pose with the native always present — a local funnel test, not blind docking.

KEY POINTS

- Affinity-only channels barely move r — but the hypothesis was about the pose
- On native-pose recognition the same channels, physically weighted, succeed strongly
- This supports the hypothesis: the gain is in using the pose, not an affinity number alone

Measured in this work (n = 250). The test is a local score-versus-RMSD funnel, stated as such rather than reported as blind docking performance.

RESULTS

It binds like this; we predict that

Crystal pose against our top-ranked predicted pose, on held-out complexes

RESULT FIGURE

figures/pose_example.png

Drop in as-is: crystal pose (navy) and top-scored predicted pose (gold) overlaid in the pocket for two held-out complexes. Chimera-ready structures in example_structures/ — protein, ligand_native (NAT), ligand_predicted (PRD)

KEY POINTS

- Crystal (true) pose and our top-ranked predicted pose, overlaid in the same pocket
- Held-out complexes; the predicted pose is picked by score alone, the native never given
- They nearly coincide: RMSD 1.16 Å and 0.95 Å

Sub-Ångström agreement on complexes the scorer never saw — this is what the pose weights buy.

Measured in this work on held-out complexes; poses selected by score with the native withheld from the selection.

RESULTS

The scoring funnel toward the native pose

Score against RMSD for three weightings of the same channels

RESULT FIGURE

figures/pose_funnel.png

Drop in as-is: score-versus-RMSD scatter for affinity-fit, contact-only and physical pose weights

KEY POINTS

- A good scorer gives its highest scores to the lowest-RMSD poses — that is the funnel
- Contact alone already funnels; physical weights sharpen it
- Affinity-fit weights do not funnel at all — the table's message, drawn

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Crude inputs
Gasteiger charges on PDB structures and name-based atom typing. Better force-field charges did not improve affinity r — an honest negative result.

Cognate redocking on PDBbind only
No external affinity test, and no enrichment (LIT-PCBA) or pose (PoseBusters) benchmark has been run yet.

None of these are hidden in the appendix: they are what the next phase of the project has to fix.

Further study — and why TBK1 matters

From a local funnel to blind docking, then to an animal-biotechnology target

NEXT STEPS

Method

Learn pose-appropriate weights rather than affinity-fit weights, then move from the local funnel to blind FFT docking.

Benchmarks

Run PoseBusters for pose quality (RMSD < 2 Å) and LIT-PCBA for enrichment (EF, BEDROC) — neither has been run yet.

Why it matters — TBK1

TBK1 is a target linked to aging and fat accumulation [1]. A faithful screen could find natural-product modulators of animal metabolism and health.

Plan

Screen natural products (COCONUT) against TBK1, recovering known binders such as betaine as a positive control [1].

The order matters: pose-appropriate weights, then blind docking, then benchmarks, then the application.

Benchmarks: PoseBusters (Buttenschoen et al., Chem. Sci. 2024); LIT-PCBA (Tran-Nguyen et al., J. Chem. Inf. Model. 2020); COCONUT (Sorokina et al., J. Cheminform. 2021). [1] TBK1 in aging / fat accumulation and betaine binding — references to be confirmed by the author.